

Successful Primary Closure of a Congenital Sternal Cleft in Infancy: A Case Report

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Introduction & Pathophysiology

Definition: Rare chest wall deformity (1 in 100,000) resulting from Failure of fusion of the paired mesenchymal sternal bars

Embryology: Occurs during 6th–9th week gestation. Failure of midline fusion results in the heart and great vessels being covered only by skin and soft tissue

Classification: Complete / Incomplete

Case Presentation

Patient: 5-month-old female

Chief complaints: Visible upper chest depression with Paradoxical movement and cardiac pulsation during crying

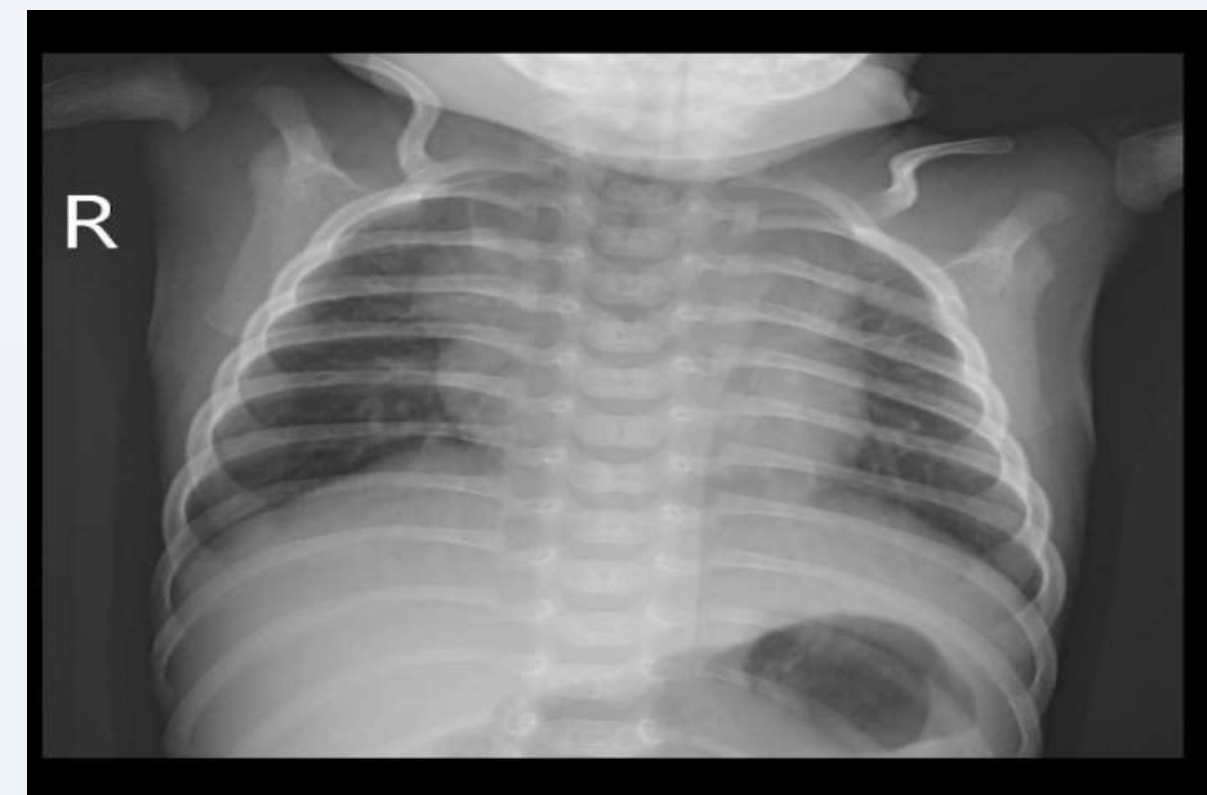
Physical Examination: Palpable V- shaped defect
No respiratory distress
Intact skin coverage

Diagnostic Imaging

CXR: Widened mediastinum

ECHO: OS-ASD, No other intracardiac anomaly

Pre-op Images



Surgical Technique

Timing: 5 months

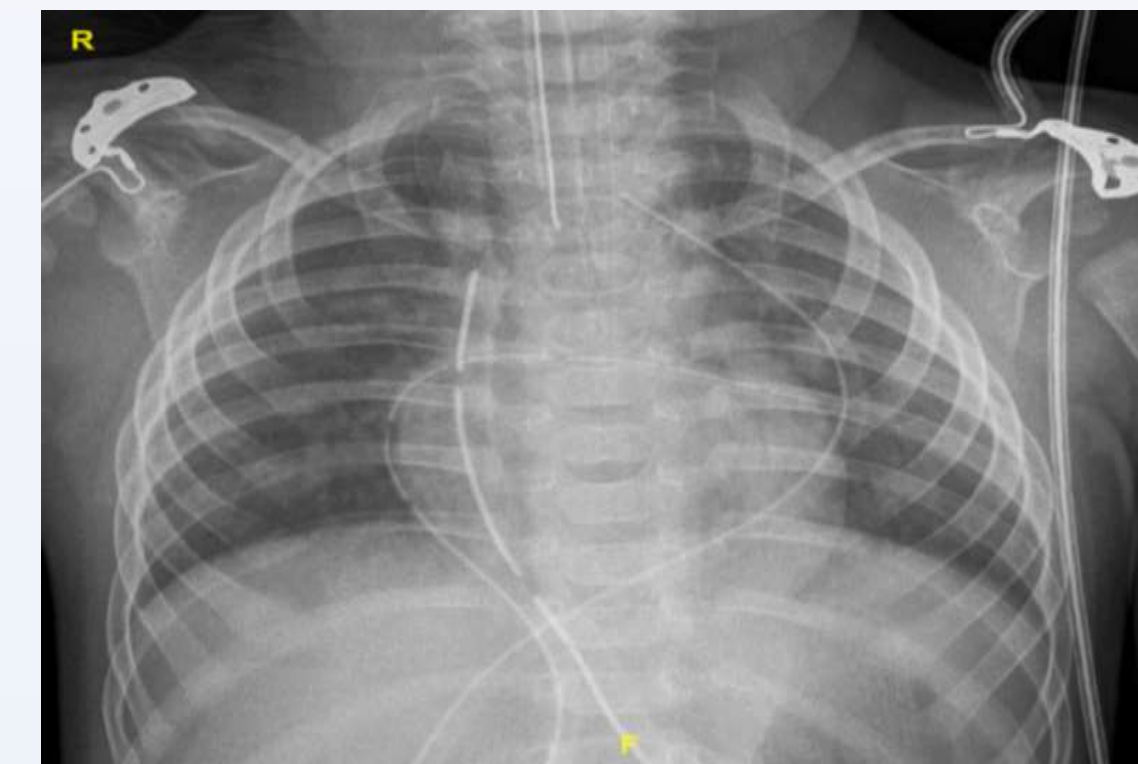
Midline incision

Mobilization of pectoralis major bilaterally

Primary closure with sutures

No chondrotomy required

Post-op Images



Postoperative Course

- ICU ventilation for 24h
- Extubated POD 1
- Discharged POD 5
- No complications

Long Term outcomes

3 months follow up

Excellent cosmetic result

Stable sternum

Normal chest wall growth

Conclusion

Rare but correctable anomaly

Early surgery preferred

Imaging essential before surgery